Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/04/2022
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (80NSSC20M0072 ; NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	04/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 04/30/2023 per D. Stockton/GRC (Ed., 6/6/22) NOTE: End date changed to 06/11/2022 per NSSC information (Ed., 5/17/22) NOTE: Extended to 4/30/2022 with new grant number 80NSSC20M0072, per NSSC information (Ed., 8/12/20) NOTE: Extended to 06/11/2020 per NSSC information (Ed., 1/29/2020) NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0072 ; NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE Update August 2020: New grant issued for Flame Design - A Novel Approach to Clean Efficient Diffusion Flames, with grant number 80NSSC20M0072, which continues to 4/30/2022. NOTE (September 2015): Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	We have completed Round 1 (normal flames) and Round 2 (inverse flames) of Flame Design testing and are processing the data. This includes making videos from HiBMs still images, converting video from the ops camera from ts to mp4 format, plotting the digital data for all tests, and measuring flame diameters. One journal paper was published in this year and three others were accepted. Five conference papers and posters were presented. Phillip Irace successfully defended his Ph.D. dissertation. Kendyl Waddell successfully presented his dissertation proposal. We continue to organize the results of the Flame Design ISS tests -- including flame diameter, fiber temperature, photomultiplier tubes (PMTs), radiometers, etc. Particular focus has been placed on using two-color ratio and ratio intensity pyrometry to obtain fiber temperature. We continue to plot and analyze the thermocouple data. We continue to process the HiBMs data to obtain soot volume fraction and are using the Advanced Combustion via Microgravity Experiments (ACME) images to obtain soot temperature. Fiber temperature is being used to measure flame temperature at the sooting limit. We have compiled all of our radiative extinction data, excluding extinction temperature. An analytical model of steady-state diameters of gas-fed spherical diffusion flames was improved and included in several journal paper submissions. Our work on the numerical model has continued. We have run multiple cases with the University of California San Diego (UCSD) mechanism. We continue to investigate the boundary conditions for radiation and conduction. We simulated ethylene flames to perform parametric studies in an attempt to find conditions that are favorable to steady state flames.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Other Journals or Periodicals	Irace PH, Waddell K, Constales D, Yablonsky G, Kim M, Sunderland PB, Axelbaum RL. "On the existence of steady state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss." Proc. Combust. Inst, in revision as of May 2022. , May-2022
Articles in Other Journals or Periodicals	Irace PH, Waddell K, Constales D, Sunderland PB, Axelbaum RL. "Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar." Proc. Combust. Inst, in revision as of May 2022. , May-2022
Articles in Peer-reviewed Journals	Wang Z, Sunderland PB, Axelbaum RL. "Double blue zones in inverse and normal laminar jet diffusion flames. " Combustion and Flame. 2020 Jan; 211:253-9. https://doi.org/10.1016/j.combustflame.2019.09.014 , Jan-2020
Articles in Peer-reviewed Journals	Irace PH, Lee H, Waddell K, Tan L, Stocker DP, Sunderland PB, Axelbaum RL. "Observations of long duration microgravity spherical diffusion flames aboard the International Space Station. " Combustion and Flame. 2021 Mar 7;229:111373. https://doi.org/10.1016/j.combustflame.2021.02.019 , Mar-2021